

Data Storage

Learning Objectives

- Identify different storage devices and their uses
- Compare primary and secondary storage
- Understand units of digital storage from bytes to gigabytes

Engage (5-7 minutes)

Think of storing grain: small jars for daily use (RAM), large bins for the season (hard disk), and sealed packets for years (cloud). Computers store data similarly in different places for different purposes.

Explore (10-12 minutes)

List all storage devices you can find or have used: pen drives, memory cards, hard disks, phones. For each, estimate how much it can hold. Create a chart comparing their sizes and capacities.

Explain (10-12 minutes)

Primary storage (RAM) is temporary and fast like a working desk. Secondary storage (hard disk, SSD) is permanent like a cupboard. Units progress: 1 Byte = 8 bits, 1 KB = 1024 Bytes, 1 MB = 1024 KB, 1 GB = 1024 MB, 1 TB = 1024 GB.

Elaborate (8-10 minutes)

Calculate storage needs: a text file is 5 KB, a photo is 3 MB, a song is 5 MB, a video is 500 MB. How many of each can fit on an 8 GB pen drive? Write your calculations and discuss which type uses the most space.

Evaluate (5-8 minutes)

Exit ticket: Name the difference between RAM and hard disk. List storage units from smallest to largest.